



Greatest Common Divisors

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Introduction

- A key part of cryptography is being able to determine the greatest common divisor of integers.
- There are various algorithms that exist but in this course we only look at one of them.
- Given any two integers a and b , not both zero, consider the set S of integers that divide both a and b .
- The set S is necessarily nonempty and finite.
- So it makes sense to speak of the greatest element of S .

Definition

- Let $a, b \in \mathbb{Z}$ with a and b not both zero. The greatest common divisor of a and b denoted $\gcd(a, b)$ is the greatest positive integer d such that $d|a, d|b$.
- If $\gcd(a, b) = 1$ then a and b are said to be relatively prime.

Example

- The divisors of 24 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12$ and ± 24
- The divisors of 60 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 10, \pm 12, \pm 15, \pm 20, \pm 30$ and ± 60
- So the greatest common divisors of 24 and 60 is 12 denoted as $\gcd(24, 60) = 12$.

Example

- The divisors of 24 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 8, \pm 12$ and ± 24
- The divisors of 60 are $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 10, \pm 12, \pm 15, \pm 20, \pm 30$ and ± 60
- The divisors of 30 are $\pm 1, \pm 2, \pm 3, \pm 5, \pm 6, \pm 10, \pm 15$ and ± 30
- So the greatest common divisors of 24, 30 and 60 is 6 which is denoted as $\gcd(24, 30, 60) = 6$.

The Euclidean Algorithm

- Our current method for finding the greatest common divisor of two integers is to list all divisors of each integer, find those divisors common to the two lists, and then choose the greatest such common divisor.
- Surely this method becomes unwieldy for large integers
- For example what is the greatest common divisor of 803 and 154?
- Is there a better method for computing the greatest common divisor of two integers?

The Euclidean Algorithm

- Is there a better method for computing the greatest common divisor of two integers?
- Yes!
- It's called the Euclidean Algorithm



The Euclidean Algorithm

- Let $a, b \in \mathbb{Z}$ with $a \geq b \geq 0$. By the division algorithm, there exists $q_1, r_1 \in \mathbb{Z}$ such that $a = bq_1 + r_1$, $0 \leq r_1 < b$
- If $r_1 > 0$, there exists (by the division algorithm) $q_2, r_2 \in \mathbb{Z}$ such that $b = r_1q_2 + r_2$, $0 \leq r_2 < r_1$
- If $r_2 > 0$, there exists (by the division algorithm) $q_3, r_3 \in \mathbb{Z}$ such that $r_1 = r_2q_3 + r_3$, $0 \leq r_3 < r_2$
- Continue this process. Then $r_n = 0$ for some n . If $n > 1$ then $\gcd(a, b) = r_{n-1}$.
- If $n = 1$ then $\gcd(a, b) = b$.

Example

- Find $\text{gcd}(803, 154)$



Example

- Express $\gcd(803, 154)$ as an integral linear combination of 803 and 154
- Essentially, work through the steps of the previous example backwards
- This is an expression of $\gcd(803, 154)$ as an integral linear combination of 803 and 154 as desired.

Notes

- Three remarks are in order here.
- The first is that the greatest common divisor of two integers is the least positive number expressible as an integral linear combination of the integers
- Second, the expression 11 as an integral linear combination of 803 and 154 is not unique.
- There are infinitely many expressions of 11 as an integral linear combination of 803 and 154.
- Finally the above algorithm is called Extended Euclidean algorithm.



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Fundamental Theorem of Arithmetic

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Introduction

- The Fundamental Theorem of Arithmetic is our first big theorem of cryptography, but not our last!
- This theorem guarantees that any integer greater than 1 can be decomposed into a product of prime numbers.
- Furthermore, this decomposition is unique except for the order in which the prime numbers are listed

Theorem P5-P6

- Theorem P5: Let $a, b, p \in \mathbb{Z}$ with p prime. If $p \mid ab$, then $p \mid a$ or $p \mid b$.
- Theorem P6: Let $a_1, a_2 \dots a_n, p \in \mathbb{Z}$ with p prime. If $p \mid a_1 a_2 \cdots a_n$ then $p \mid a_i$ for some i .

Theorem P7

- Every integer greater than 1 can be expressed in the form $p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_n^{a_n}$ with $p_1, p_2, \dots, p_n$ distinct prime numbers and $a_1, a_2, \dots, a_n$ positive integers.
- This prime factorisation is unique except for the arrangement of the $p_i^{a_i}$.

Example

- Find the prime factorisation of 756.
- We have $756 =$



Lowest common multiple

- Let $a, b \in \mathbb{Z}$ with $a, b > 0$. The least common multiple of a and b denoted $\text{lcm}(a, b)$ is the least positive integer m such that $a|m$ and $b|m$.



Example

- The positive multiples of 6 are 6, 12, 18, 24, 30, 36, 42, 48...
- The positive multiples of 7 are, 7, 14, 21, 28, 35, 42, 49, ...
- It is a fact that the positive common multiples of 6 and 7 are 42, 84, 126, 168, 210, ...
- Obviously from these common multiples of 6 and 7 the least common multiple of 6 and 7 is 42, which is denoted as $\text{lcm}(6, 7) = 42$.

Example

- The positive multiples of 6 are 6, 12, 18, 24, 30, 36, 42, 48...
- The positive multiples of 8 are, 8, 16, 24, 32, 40, 48, 56, ...
- From the common multiples of 6 it is obvious the least common multiple of 6 and 8 denoted as $\text{lcm}(6, 8) = 24$.

Notes

- Instead of just listing the possible number of multiples we can use the fundamental theorem of Arithmetic to compute both:
 - Greatest common divisor
 - Lowest common multiple

Example

- Find $\text{gcd}(756, 2205)$ and $\text{lcm}(756, 2205)$ by using prime factorisation.



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Congruences

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Introduction

- Congruences play a vital role in cryptography
- In-fact thanks to congruences we have very strong public key cryptographic systems
- The notion of congruence is defined in terms of divisibility relation.
- Two integers are related under congruence if their difference is evenly divisible by a third fixed positive integer as now specified.

Definition

- Let $a, b, m \in \mathbb{Z}$ with $m > 0$. Then a is said to be congruent to b modulo m , denoted $a \equiv b \pmod{m}$ if $m \mid a - b$.
- If $a \equiv b \pmod{m}$ then m is said to be the modulus of the congruence.
- The notation $a \not\equiv b \pmod{m}$ means that a is not congruent to b modulo m ; a is said to be incongruent to b modulo m .

Example

- Since $4 \mid 25 - 1$, we have $25 \equiv 1 \pmod{4}$
- Since $6 \mid 4 - 10$, we have $4 \equiv 10 \pmod{6}$
- Since $7 \mid 10 - (-4)$, we have $10 \equiv -4 \pmod{7}$
- Since $1 \mid a - b$, we have $a \equiv b \pmod{1}$
- Since $5 \nmid -7 - 2$ we have $-7 \not\equiv 2 \pmod{5}$

Equivalence Classes

- We now undertake an analysis of the structure of the equivalence classes of the integers under congruence modulo m .
- Throughout our discussion, we use the notation $[x]$ to denote the equivalence class containing x .
- Our analysis begins with a motivational example

Example

- The equivalence classes of $\mathbb{Z}$ under congruence modulo 4 are



Notes

- A set of integers such that every integer is congruent modulo m to exactly one integer of the set is said to be a complete residue system modulo m .
- Example: $\{0, 1, 2, 3\}$ is a complete residue system modulo 4
- $\{6, -11, 19, 1988\}$ is a complete residue system modulo 4.
 - $6 \equiv 2 \pmod{4}$
 - $-11 \equiv 1 \pmod{4}$
 - $19 \equiv 3 \pmod{4}$
 - $1988 \equiv 0 \pmod{4}$



Practice problems 1

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Example 1

- Prove or disprove each statement below
 - i) $6|42$
 - ii) $4|50$
 - iii) $16|0$
 - iv) $0|15$

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Example 2

- Find the unique integers q and r guaranteed by the division algorithm with each dividend and divisor below
 - i) $a=47$ $b=6$
 - ii) $a=281$ $b=13$
 - iii) $a=343$ $b=49$

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Example 3

- Determine whether the following positive integers are prime or composite
 - i) 127
 - ii) 129
 - iii) 131



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Practice problems 2

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Example 1

- Find the greatest common divisor of
 - $\text{gcd}(37, 60)$
 - $\text{gcd}(78, 708)$
 - $\text{gcd}(441, 1155)$





Practice problems 3

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Example 1

- Find the prime factorisation of each of the integers below
 - i) 51
 - ii) 87
 - iii) 361
 - iv) 367



Example 2

- Find the least common multiple of
 - i) $\text{lcm}(6,9)$
 - ii) $\text{lcm}(13,91)$
 - iii) $\text{lcm}(100,105)$



Example 3

- Find the gcd using prime factorisation of the following set of numbers
 - i) $\text{gcd}(1,260, 8,820)$





Integer Rings

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Introduction

- After studying the properties of modulo reduction we are now ready to define in more general terms a structure that is based on modulo arithmetic.
- Let's look at the mathematical construction that we obtain if we consider the set of integers from zero to $m-1$ together with the operations addition and multiplication:

Definition

- The integer ring $\mathbb{Z}_m$ consists of:
 - The set $\mathbb{Z}_m = \{0, 1, 2, 3, \dots, m - 1\}$
 - Two operations $+$ and $\times$ for all $a, b \in \mathbb{Z}_m$ such that:
 - $a + b \equiv c \pmod{m} \quad c \in \mathbb{Z}_m$
 - $ab \equiv d \pmod{m} \quad d \in \mathbb{Z}_m$

Example

- Let $m=9$, so we are dealing with the ring $\mathbb{Z}_9 = \{0,1,2,3,4,5,6,7,8\}$
- Let's look at a few simple arithmetic operations:
- $6+8=14 \equiv 5 \pmod{9}$
- $6*8=48 \equiv 3 \pmod{9}$

Notes

- We will discuss more about rings and finite fields later on but for now the following properties of rings are important:
- We can add and multiply any two numbers and the result is always in the ring. Therefore a ring is closed
- Addition and multiplication are associative, e.g.
 $a+(b+c)=(a+b)+c$ and $a*(b*c)=(a*b)*c$ for all $a, b, c \in \mathbb{Z}_m$

Notes

- There is a neutral element 0 with respect to addition, i.e., for every element $a \in \mathbb{Z}_m$ it holds that $a+0 \equiv a \pmod{m}$
- For any element a in the ring, there is always the negative element $-a$ such that $a+(-a) \equiv 0 \pmod{m}$ (additive inverse always exists)
- There is the neutral element 1 with respect to multiplication, i.e. for every element $a \in \mathbb{Z}_m$ it holds that $a*1 \equiv a \pmod{m}$

Notes

- The multiplicative inverse exists only for some, but not for all elements.
- Let $a \in \mathbb{Z}_m$ then the inverse a^{-1} is defined such that $a \cdot a^{-1} \equiv 1 \pmod{m}$
- It takes some effort to *find* the inverse! We shall see the method soon

Notes

- However, there is an easy way of telling whether an inverse for a given element a exists or not.
- An element $a \in \mathbb{Z}$ has a multiplicative inverse a^{-1} if and only if $\gcd(a, m) = 1$, where $\gcd$ is the greatest common divisor.
- The fact that two numbers have a $\gcd$ of 1 is of great importance in number theory, and there is a special name for it: if $\gcd(a, m) = 1$, then a and m are said to be *relatively prime* or *coprime*

Example

- Let's see whether the multiplicative inverse of 15 exists in $\mathbb{Z}_{26}$
- Since $\gcd(15, 26) = 1$ the inverse must exist.
- On the other hand, since $\gcd(14, 26) \neq 1$ the multiplicative inverse of 14 does not exist in $\mathbb{Z}_{26}$

Notes

- Another ring property is that $a*(b+c)=(a*b)+(b*c)$ for all $a, b, c \in \mathbb{Z}_m$, i.e., the distributive law holds.
- In summary, roughly speaking, we can say that the ring $\mathbb{Z}_m$ is the set of integers $\{0, 1, 2, \dots, m-1\}$ in which we can add, subtract, multiply and sometimes divide.

Notes

- As mentioned earlier, the ring $\mathbb{Z}_m$ and thus integer arithmetic with the modulo operation, is of central importance to modern public-key cryptography.
- In practice, the integers involved have a length of 150–4096 bits so that efficient modular computations are a crucial aspect.

Notes

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Shift Cipher

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Introduction

- We now introduce another historical cipher, the shift cipher.
- It is actually a special case of the substitution cipher and has a very elegant mathematical description.
- The shift cipher itself is extremely simple: We simply shift every plaintext letter by a fixed number of positions in the alphabet.

Introduction

- For instance, if we shift by 3 positions, A would be substituted by D.
- B by E, etc..
- The only problem arises towards the end of the alphabet: What should we do with X,Y,Z?
- As you might have guessed, they should “wrap around”.

Introduction

- That means X should become A, Y should become B and Z is replaced by a C.
- Allegedly, Julius Caesar used this cipher with a three-position shift.
- The shift cipher also has an elegant description using modular arithmetic.

Definition

- For the mathematical statement of the cipher, the letters of the alphabet are encoded as numbers, as shown by the table:



A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Definition

- Both the plaintext letters and the ciphertext letters are now elements of the ring $\mathbb{Z}_{26}$.
- Also, the key, i.e., the number of shift positions, is also in $\mathbb{Z}_{26}$ since more than 26 shifts would not make sense.

Definition

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Definition

Let $x, y, k \in \mathbb{Z}_{26}$.

Encryption: $e_k(x) \equiv x + k \pmod{26}$.

Decryption: $d_k(y) \equiv y - k \pmod{26}$.

Example

- Let the key be $k=17$ and the plaintext is ATTACK
- Let $A=x_1$ $T=x_2$ $T=x_3$ $A=T=x_4$ $C=T=x_5$ $K=T=x_6$
- Then we have the following:

Notes

- As you can guess from the discussion of the substitution cipher earlier in this course, the shift cipher is not secure at all.
- There are two ways of attacking it:
 - 1. Since there are only 26 different keys (shift positions), one can easily launch a brute-force attack by trying to decrypt a given ciphertext with all possible 26 keys.
 - If the resulting plaintext is readable text, you have found the key.

Notes

- 2. As for the substitution cipher, one can also use letter frequency analysis.





Affine Cipher

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Introduction

- Now we try to improve the shift cipher by generalising the encryption function.
- Recall that the actual encryption of the shift cipher was the addition of the key $y_i = x_i + k \bmod 26$
- The affine cipher encrypts by multiplying the plaintext by one part of the key followed by the addition of another part of the key.

Definition

Let $x, y, a, b \in \mathbb{Z}_{26}$

Encryption: $e_k(x) = y \equiv a \cdot x + b \pmod{26}.$

Decryption: $d_k(y) = x \equiv a^{-1} \cdot (y - b) \pmod{26}.$

with the key: $k = (a, b)$, which has the restriction: $\gcd(a, 26) = 1$.

Notes

- The decryption is easily derived from the encryption function:



- The restriction $\gcd(a, 26)=1$ stems from the fact that the key parameter a needs to be inverted for decryption.



Notes

- Recalling from previous classes that an element a and the modulus must be relatively prime for the inverse of a to exist.
- Thus, a must be in the set:
 - $a \in \{1,3,5,7,9,11,13,15,17,19,21,23,25\}$
- But, how do we find a^{-1} ? For now, we can simply compute it by trial and error

Notes

- For a given a we simply try all the possible values a^{-1} until we obtain: $aa^{-1} \equiv 1 \pmod{26}$
- For instance, if $a=3$, then $a^{-1} = 9$ since $3*9=27 \equiv 1 \pmod{26}$

Example

- Let the key be $k=(a,b)=(9,13)$ and the plaintext be ATTACK
- Let $A=x_1$ $T=x_2$ $T=x_3$ $A=T=x_4$ $C=T=x_5$ $K=T=x_6$

Question

- Is the affine cipher secure?
- No! The key space is only a bit larger than in the case of the shift cipher:
 - Key-space=(number of values of a)*(number of values of b) = $12 \times 26 = 312$
- A key-space with 312 elements can, of course, still be searched exhaustively, i.e., brute-force attacked, in a fraction of a second with current desktop PCs.

Notes

- In addition, the affine cipher has the same weakness as the shift and substitution cipher: The mapping between plaintext letters and ciphertext letters is fixed.
- Hence, it can easily be broken with letter frequency analysis.
- The remainder of this course deals with strong cryptographic algorithms which are of practical relevance.